



OPEN Finite element analysis of chloride ion penetration and service life prediction in concrete with supplementary cementitious materials

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This study investigates chloride ion penetration and service life (SL) prediction in concrete, highlighting the role of supplementary cementitious materials (SCMs), such as silica fume (SF) and ground granulated blast furnace slag (GGBFS). Finite element (FE) simulations using COMSOL Multiphysics were performed with both constant and time-dependent chloride diffusion coefficients obtained from experimental data. Four concrete mixtures with varying water-to-binder (w/b) ratios and SCM contents were tested. Chloride diffusion coefficients, measured from samples exposed to a 165 g/L NaCl solution for 3, 6, 9, and 12 months, were incorporated into the simulations. The results showed that concretes with a higher w/b ratio (0.55) experienced greater chloride penetration, with diffusion coefficients ranging from $8.574 \times 10^{-12} \text{ m}^2/\text{s}$ (at 3 months) to $5.635 \times 10^{-12} \text{ m}^2/\text{s}$ (at 12 months), whereas concretes with a lower w/b ratio (0.45) exhibited superior resistance, with diffusion coefficients decreasing from $6.494 \times 10^{-12} \text{ m}^2/\text{s}$ (at 3 months) to $4.401 \times 10^{-12} \text{ m}^2/\text{s}$ (at 12 months). SCMs significantly reduced chloride penetration in both w/b ratios; SF-containing concrete achieved the lowest diffusion coefficient ($1.563 \times 10^{-12} \text{ m}^2/\text{s}$ at 12 months), due to its ability to densify the concrete matrix. Time-dependent diffusion modelling improved SL prediction accuracy, reducing simulation errors to below 30%, compared to 48–75% for constant models. SL predictions further highlighted the durability benefits of SCMs: concrete with GGBFS demonstrated a potential SL of 62 years (variable diffusion model), compared to 10 years (constant model), while SF-enhanced concrete achieved even greater longevity, with SL extending up to 106 years using variable models. These findings underscore the critical importance of SCMs, appropriate w/b ratios, and dynamic diffusion modelling in enhancing durability and ensuring the longevity of infrastructure in chloride-rich environments.

Keywords COMSOL, Finite element simulation, Ground granulated blast furnace slag, Silica fume, Constant and variable chloride ions diffusion coefficient

Concrete, a fundamental material in civil engineering, is renowned for its versatility, durability, and widespread application in construction. Reinforcement in concrete plays a crucial role in enhancing the structural capacity and longevity of concrete elements, particularly in large-scale infrastructure projects. The integration of steel reinforcement helps counteract the inherent weakness of concrete in tension, thereby improving its performance under various loading conditions over time. However, the presence of steel reinforcement also introduces potential durability concerns, particularly the risk of corrosion over time.

Reinforcement corrosion, induced by chloride ion ingress from environmental sources such as seawater, deicing salts, and marine atmospheres, poses a significant threat to the durability of reinforced concrete structures (RCSs). It is considered one of the most common causes of deterioration. Two primary mechanisms protect steel reinforcements embedded within concrete. First, physical protection is provided by the concrete cover, which acts as a barrier to chloride ions; the depth of penetration increases proportionally with concrete porosity¹. Second, chemical protection arises from the high alkalinity of concrete, which forms a passive oxide

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layer on the steel surface, shielding it from corrosion². Despite these mechanisms, chloride ions can penetrate the concrete over time, compromising the protective layers and initiating the corrosion process.

However, due to the inherent porosity of concrete, chloride ions, along with moisture and oxygen, gradually infiltrate over time. This compromises the protective oxide layer, accelerating the corrosion process^{2–7}. The expansion of corrosion products further exacerbates the situation, inducing internal stresses that contribute to concrete cracking^{6,8}. Moreover, corrosion-related deterioration can lead to the separation and cracking of the concrete cover, which decreases the reinforcement's effective cross-sectional area and weakens its bond with the concrete matrix^{3,7,9}.

As corrosion persists, the structural integrity of concrete is progressively compromised, potentially resulting in severe loss of serviceability and significant reduction in load-bearing capacity, sometimes leading to sudden structural failure⁴. Given these critical implications, addressing reinforcement corrosion and ensuring long-term concrete durability remain paramount challenges in civil engineering practice^{7,10}.

Numerous factors influence the durability of CSs, including environmental conditions, material quality, construction practices, and maintenance protocols. Ensuring durable concrete requires meticulous attention to various measures, including optimal mix design, the selection of appropriate materials, precise batching and mixing procedures, effective transportation and placement practices, and adequate curing techniques.

Furthermore, alongside strategies aimed at enhancing the concrete durability, it is imperative to assess the resistance of concrete to environmental challenges initially. Numerous researchers have investigated chloride ion penetration in concrete^{11–15} and used Fick's second law (Eq. (1)), as outlined below¹⁶:

$$\frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial x^2} \quad (1)$$

In this equation, C signifies the concentration of chloride, D denotes the diffusion coefficient, x is the distance from the surface of the concrete and t represents time¹⁶. Many researchers have conducted studies to apply Fick's second law for estimating the SL of RCSs¹⁷. Collepardi et al.¹⁷ utilised the mathematical solution to this equation, which provided by Crank¹⁸, to estimate the chloride ions diffusion (refer to Eq. (2)).

$$\frac{C}{C_s} = \operatorname{erfc}\left(\frac{x}{2\sqrt{D_t \cdot t}}\right) \quad (2)$$

Where C_s and erfc denote the surface concentration of chloride and the error function, respectively. The D_t is a crucial parameter for assessing the chloride penetration into concrete. A higher D represents greater ease of chloride ion penetration¹⁶. It has been noted that the value of D decreases as curing age and time progress, a phenomenon particularly pronounced in the early stages^{16,19,20}. Given the significant impact of the time-dependent reduction in the D on the prediction of SL, it must be accounted for modelling²⁰. The time-dependent D can be described by an exponential relationship, where D at a reference time is used as the basis. Numerous researchers have presented Eq. (3) to investigate the temporal variations in Dt ^{16,20–22}:

$$D_t = D_0 \left(\frac{t_0}{t}\right)^m \quad (3)$$

Where m represents the time-dependent index and D_0 represents the diffusion coefficient at a specified age t_0 . Different studies have established the value of m and provided corresponding values according to pozzolan content, ratio, and mix design^{16,20–22}.

Following the analysis of penetration mechanisms in concrete, accurately predicting the SL of various concrete mixes becomes essential. Such predictions help select optimal mix designs and define maintenance intervals and costs²⁰. Several definitions of SL have been proposed in the literature, each emphasising different durability criteria. Zhang and Ba²³ define SL as the time until chloride ions reach a critical threshold near the reinforcement. Chai et al.²⁴ and Vořechovská et al.²⁵ consider the breakdown of the passive layer as indicative of SL, while Parameswaran et al.²⁶ describe it as the period during which a structure maintains its efficiency. Martin-Perez and Lounis²⁷ define SL as the time until damage becomes unacceptable, and Abu-Tair et al.²⁸ define it as the period until significant damage requiring costly repairs occurs. Těplý²⁹ refers to SL as the period during which the structure can operate under intended conditions. Although numerous studies have focused on SL prediction of concrete structures^{30–36}. Challenges remain due to the heterogeneous microstructure of concrete¹¹. A critical parameter in estimating SL is the chloride threshold level (CTL)^{10,34}. This represents the chloride concentration at the steel surface needed to initiate corrosion. For accurate SL prediction, the total chloride content near the reinforcement must exceed the CTL^{34,37}.

As previously noted, the CTL is a crucial factor in modelling the SL of RCSs. Nevertheless, there is presently no widely accepted approach for establishing the CTL³⁸. Accurately assessing the chloride concentration that initiates corrosion on the surface of reinforcements requires meticulous examination. The CTL is affected by several factors, including the amount and type of binder, the reinforcement's electrochemical potential, the chloride binding capacity, the hydroxyl ion concentration, the presence of oxygen, the surface condition of the reinforcement, and temperature^{10,38}. Consequently, the CTL is typically specific to each concrete type and its exposure conditions³⁸. Nonetheless, past studies have commonly reported CTL values of 0.4–0.8 w.t% of cement/binder^{10,34,38–42} or 0.05–0.1 w.t% of concrete^{20,39–41,43}.

Extensive research has been conducted on simulating chloride penetration and predicting the SL of RCSs using various FE software tools. These tools represent a significant advancement in durability design by providing predictive models to estimate SL³⁴. Generally, if simulation models are well-calibrated against experimental data,

they can accurately and efficiently predict the long-term behaviour of concrete structures^{19,44}. Furthermore, simulations can reveal phenomena not captured in experiments, offering more profound insight into chloride transport mechanisms^{11,19}. Life-365 is one of the most widely used tools in this domain, applied over the past 15 years to estimate the SL of concrete exposed to chlorides⁴⁵. COMSOL Multiphysics is another advanced software that offers a broad range of simulation capabilities. It solves differential equations for nonlinear systems in one, two, or three dimensions using the FE method and excels at integrating multiple physical phenomena into unified models. Originally developed at the Royal Institute of Technology in Sweden⁴, COMSOL has demonstrated high accuracy when validated against experimental data and MATLAB results^{4,11,19,46}. For example, studies have shown that using COMSOL to simulate chloride transport and binding in cracked concrete provides valuable insights, confirming its utility for assessing chemical ingress^{11,47}.

Two critical parameters influencing the durability of RCSs against chloride ion attack, and consequently the corrosion of the reinforcement, are the w/b ratio and the incorporation of SCMs. The w/b ratio directly affects the pore structure and permeability of concrete, which are key factors in chloride ingress and subsequent corrosion^{48,49}. A lower w/b ratio results in a denser matrix with fewer and smaller pores, thus reducing permeability and improving resistance to chloride-induced corrosion^{49–51}. Conversely, a higher w/b ratio increases porosity, facilitating chloride penetration and thereby increasing the corrosion risk. Badalyan et al.⁵² demonstrated that lower w/b ratios significantly enhance durability, while Mehta and Monteiro⁵³ noted that high w/b ratios make concrete more susceptible to deterioration. SCMs, such as SF and GGBFS, also play a crucial role in enhancing resistance by modifying the microstructure and chemical properties of concrete. SF, a by-product of silicon and ferrosilicon production, exhibits high pozzolanic reactivity and reacts with calcium hydroxide to form additional C-S-H gel, refining pores and reducing permeability^{54–56}. GGBFS, a by-product of iron and steel manufacturing, is another SCM that contributes to improved durability. Similarly, GGBFS, a byproduct of steel production, reacts with the alkaline components of cement to produce more C-S-H, thereby enhancing chloride binding and reducing permeability. Recent work has confirmed that GGBFS incorporation lowers chloride diffusion and improves resistance by forming a stable, less porous microstructure^{46,57–59}.

Previous studies have employed various methods, including Fick's second law and FE simulations, to model chloride ion diffusion in concrete. Recent studies^{60–62} have demonstrated that incorporating SCMs, along with time-dependent diffusion and surface chloride concentration, significantly enhances the accuracy of SL predictions for concrete exposed to chlorides. Although some researchers have utilised FE software, such as COMSOL Multiphysics, for these simulations, most of these studies assume a constant chloride diffusion coefficient, failing to account for the time-dependent evolution of concrete's microstructure and permeability. While such models provide valuable insights into initial chloride ingress, they fall short in accurately capturing the long-term behaviour of concrete, where diffusion properties are influenced by ongoing hydration, pore refinement, and the beneficial effects of SCMs, such as SF and GGBFS.

The research gap lies in the lack of comprehensive studies that integrate time-dependent chloride diffusion coefficients with the effects of SCMs in numerical simulations, particularly for predicting the SL of RCSs. Most existing models neither incorporate experimental data for variable diffusion coefficients nor consider the combined effects of SCMs over extended exposure periods. Furthermore, there is limited research employing COMSOL Multiphysics to simulate chloride diffusion under realistic conditions, accounting for these critical variables.

This study addresses these gaps by utilising COMSOL Multiphysics to simulate chloride ion penetration using both constant and time-dependent diffusion models. The simulations are informed by experimental data on chloride diffusion coefficients, measured from concrete specimens exposed to a 165 g/L NaCl solution for 3, 6, 9, and 12 months. By explicitly incorporating different SCMs (SF and GGBFS), this research systematically evaluates their effects on chloride resistance and develops a more accurate framework for SL prediction in concrete structures. Unlike previous studies that relied on simplified assumptions, this work demonstrates the importance of combining time-dependent diffusion behaviour with the influence of SCMs to improve the prediction of long-term durability in chloride-laden environments.

Research program

The workflow depicted in Fig. 1 outlines a systematic approach to investigating chloride ion penetration in concrete, using both experimental and numerical methods. The process begins with Step 1: Mixture Design, where four concrete mixtures are prepared. These mixtures vary in w/b ratios and the inclusion of SCMs. The mixtures include OPC with w/b ratios of 0.45 and 0.55, as well as mixtures containing SF and GGBFS, each with a w/b ratio of 0.45. This mixture design allows for the analysis of how different compositions affect chloride ion diffusion and overall concrete durability.

In Step 2: Experimental Program, the compressive strength of the concrete is tested at 28 days, followed by chloride ion penetration tests at 3, 6, 9, 12 and 24 months. These tests provide essential empirical data on how chloride concentrations change over time and depth within the concrete, serving as a foundation for further analysis. In Step 3, the *D* is calculated using two models. In Step 3.1, a constant *D* is calculated for the 3, 6, 9, and 12-month time points using MATLAB's Curve Fitting Toolbox. In contrast, Step 3.2 calculates a variable *D* using a time-dependent equation that accounts for changes in diffusion rates over time, which is crucial for simulating more realistic chloride ingress behaviour.

Next, in Step 4: FE Simulation, the calculated *D*s are applied in numerical models to simulate chloride ion penetration through the concrete. This simulation allows for the visualisation and prediction of chloride ion concentration profiles over time. Step 5: Validation compares the simulation results with experimental data at 24 months of age, using the average error formula to quantify the model's accuracy. Finally, in Step 6: SL Prediction, the validated models are used to estimate the concrete's SL based on a chloride threshold limit (0.4% by weight of

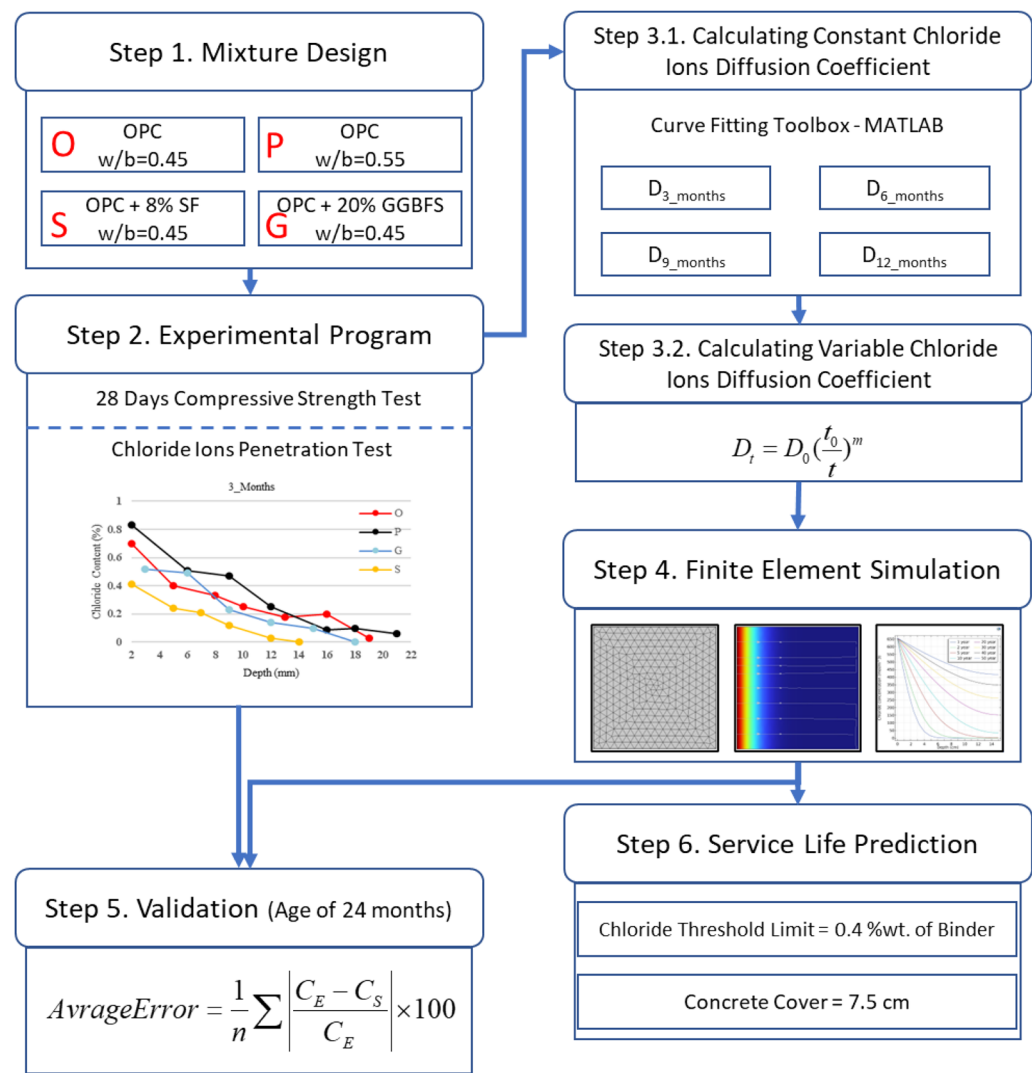


Fig. 1. The research framework.

Binders	Chemical composition (% by mass)									Physical Properties		
	CaO	SiO ₂	Al ₂ O ₃	Fe ₂ O ₃	SO ₃	MgO	Na ₂ O	K ₂ O	L.O.I	Specific Gravity	Fineness (cm ² /gr)	Retained on 45 μm Sieve (%)
OPC	65.65	22.28	4.71	2.85	2.56	1.95	0.16	0.48	1.41	3.08	3170	9.54
GGBFS	30.28	35.56	8.68	0.64	0.63	8.89	0.57	0.96	2.23	2.85	3035	18.40
SF	0.49	93.54	0.43	0.94	0.71	1.00	0.82	1.04	0.67	2.16	342,200	5.11

Table 1. Physical and chemical properties of binders.

the binder) and a predefined concrete cover of 7.5 cm. This step is critical for determining how long the concrete will remain resistant to chloride-induced corrosion under real-world conditions.

Experimental program

Materials

In this study, OPC, SF, and GGBFS were used as binders in the concrete samples, with their physical and chemical properties detailed in Table 1. The cement had a compressive strength of 35.8 MPa at 28 days, as measured on standard mortar cubes in accordance with ASTM C109. The strength activity index of the SCMs was approximately 77% for GGBFS and about 115% for SF, indicating their compliance with ASTM standards and confirming their effectiveness in enhancing the mechanical and durability properties of concrete. Additionally, crushed sand and gravel, conforming to ASTM C33⁶³ for granularity, were utilised. The fine aggregate had a fineness modulus of 2.68 and a water absorption of 3.2%, while the coarse aggregate had a water absorption of 2.55%. To achieve

workability due to the high fineness of the particles, a polycarboxylate ether-based superplasticiser (SP) with a specific gravity of 1.05 was added to the samples. The SP had a pH of 7 ± 1 and a solid content of approximately 35%.

Mixture proportions

During the experimental phase of this research, four types of concrete mixtures, labelled O, P, G, and S, were designed to evaluate the effects of the w/b ratio and the incorporation of SCMs, such as SF and GGBFS, on chloride ion penetration and its FE simulation. The specifications of these concrete mixtures are presented in Table 2. In mixtures O and P, the w/b ratios were set at 0.45 and 0.55, respectively, to facilitate comparison. For mixture S, based on the optimal percentage of SF identified in previous research⁶⁴, 8% by weight of SF was used as a cement substitute. Similarly, mixture G incorporated 20% by weight of GGBFS as a cement replacement, following recommendations from a prior study⁶⁵.

Mixing and specimen Preparation

The concrete samples were prepared following the ASTM C192⁶⁶. For all specimens, a SP was added to a portion of the mixing water to enhance workability. The freshly mixed concrete was then placed into 100 and 150 mm cubic moulds and cured for 24 h at ambient temperature in a humidity chamber. After curing, the samples were demolded and transferred to a chamber with saturated lime water maintained at 21 ± 3 °C, where they remained until the time of testing. For each mixture and each testing age, three specimens were prepared and tested, and the average results were reported.

Test methods

Compressive strength test A 2000 kN concrete compression testing equipment was used to determine the compressive strength of $100 \times 100 \times 100$ mm concrete specimens at a loading rate of $0.5 \text{ N.mm}^{-2}.\text{s}^{-1}$. This experiment was conducted after 28 days of curing on three samples of each mixture.

Chloride ions penetration test In a thoroughly structured experimental program, 150 mm cubic samples were prepared using four distinct concrete mixtures. After curing in a moist environment for 7 days, epoxy resin was applied to all but one surface of each sample, allowing unidirectional chloride penetration. The specimens were then immersed in a 165 g/L NaCl solution and left to soak for 2 years⁶⁷. At the testing stage, once the specimens were dried, powder samples were collected by drilling successive layers of approximately 2 mm thickness from the exposed surface. The extracted powders were mixed with nitric acid and distilled water to prepare solutions based on ASTM C1152⁶⁸. These solutions were subsequently heated, filtered, and analysed by potentiometric titration to determine the chloride content, following ASTM C1152. Figure 2 illustrates the experimental procedure. It is important to note that while chloride profile testing was performed at 3, 6, 9, and 12 months for modelling purposes, additional data at 24 months were also collected and used solely for validating the simulation results, as discussed in Sect. 4. This procedure was repeated at least five times to assess the chloride ion penetration profile of each specimen at various exposure ages (3, 6, 9, 12, and 24 months). Finally, MATLAB software was utilised to calculate the diffusion coefficient (D), facilitating the development of a numerical model. Several approaches have been proposed for estimating the time-dependent diffusion coefficient, including regression analysis of experimental data or adopting fixed m values as suggested in previous studies. While the regression-based integrated approach — which simultaneously fits $D(t)$ and $C_s(t)$ into a unified model⁶² — is considered rigorous and comprehensive, Eq. (3) was employed in this study with m values reported in the literature^{16,22,69}, specifically 0.62, 0.30, and 0.71 for the S, O&P, and G mixtures, respectively, under submerged conditions. This decision was due to the limited number and range of experimental data at different ages, which constrained the reliability of the regression. Furthermore, the m-based method remains a well-established and widely accepted approach in the literature for comparative studies.

FE simulation program

For a comprehensive long-term assessment, laboratory test outcomes were incorporated as input factors for the computational simulation. COMSOL Multiphysics software was employed for this task. This software is particularly useful for modelling various durability and mechanical-related factors in RCSs due to its diverse modules. In this project, several key inputs for the COMSOL model included the bulk diffusion coefficient, surface chloride concentration, geometry of the cross-section, sealant properties, and time-varying changes in the D. These parameters are discussed in more detail below. Additionally, based on the specific simulation conditions, further detailed information, such as variations in moisture and temperature content, sorption rates, diffusion coefficients within voids and concrete porosity can be provided to the software.

Concrete Mixes	OPC (kg/m ³)	w/b	GGBFS (kg/m ³)	SF (kg/m ³)	Water (kg/m ³)	Gravel (kg/m ³)	Sand (kg/m ³)	SP (% of binders)	Slump (cm)	28d_CS (MPa)
O	325	0.45	0	0	146.25	627.5	1192.7	0.39	10	50.9
P	325	0.55	0	0	178.75	628.2	1139.4	0.10	12	36.2
G	260	0.45	65	0	146.25	656.0	1189.9	0.33	13	38.2
S	299	0.45	0	26	146.25	654.2	1186.8	0.77	11	51.8

Table 2. Physical and chemical properties of binders.



Fig. 2. Test setup and procedure of the chloride ions penetration test.

In COMSOL, the transport of chemical species is modelled by solving transport equations that account for both convection and diffusion processes. The equation for the conservation of mass for chemical species is stated as follows:

$$\frac{\partial c_i}{\partial t} + \nabla \cdot J_i + u \cdot \nabla c_i = R_i \quad (4)$$

Where R_i denotes the reaction rate ($\text{mol}/(\text{m}^2 \cdot \text{s})$), D_i is the diffusion coefficient (m^2/s), C_i represents the species concentration (mol/m^3) and u is the mass-averaged velocity vector (m/s).

This equation accounts for diffusion and convection as the primary transport mechanisms. The relative mass flux J_i ($\text{mol}/(\text{m}^2 \cdot \text{s})$) is applied in boundary condition calculations and flux analysis and linked to the mass balance equation. The Transport of Diluted Species interface inherently handles molecular diffusion, where the diffusive flux J_i is defined as per Eq. (5).

$$J_i = -D_i \cdot \nabla c_i \quad (5)$$

The model allows for direct input of the diffusion coefficient, including the option to specify an anisotropic diffusion tensor. The convective term, which depends on velocity u , can be defined analytically or derived by coupling the model with a fluid flow interface, such as Laminar Flow. The right portion of Eq. (5) includes R_i , which accounts for sources or sinks, generally caused by chemical reactions or desorption from a porous medium. The Transport of Diluted Species interface also supports additional nodes, such as the Reaction node, which allows for the specification of reaction terms using the variable names of the species involved. In cases where the fluid is incompressible, the term $u \cdot \nabla c$ becomes zero, leading to the simplified conservative formulation.

When modelling in COMSOL Multiphysics, the problem and its environmental conditions must first be defined. The software supports simulations in one, two, or three dimensions. The next step involves determining the physics of the problem, which may include structural mechanics, fluid dynamics, electrochemistry, chemical species transport or solid mechanics. Afterwards, the analysis type is selected within the Study section. Following the definition of relevant units, parameters, and functions, the problem's geometry is configured. The dimensions, shapes, and coordinates are defined in Geometry. Materials are allocated in the Materials section, either from the software's library or manually. Boundary conditions, model type, and initial conditions are specified within the Solid Mechanics section. The model is then meshed, either automatically or manually. To perform the analysis, the settings within the Study section, including linearity, non-linearity, physics selection, and boundary conditions, must be finalised. The simulation is then executed, and results such as graphs and visualisations can be generated in the Results section.

In this research, a 2D model was used to simulate chloride ion penetration from the concrete surface (Fig. 3). The Transport of Diluted Species physics module was applied to model the diffusion process, with a time-dependent analysis approach. Laboratory results, including diffusion coefficients and surface chloride concentrations, were entered as input data into the software. The modelled geometry consisted of a cross-sectional square column with sides of 15 cm. The mesh consisted of 6,282 domain elements and 200 boundary

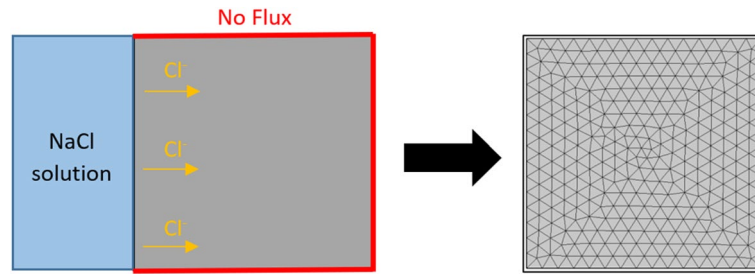


Fig. 3. Geometry schematic and meshing of models.

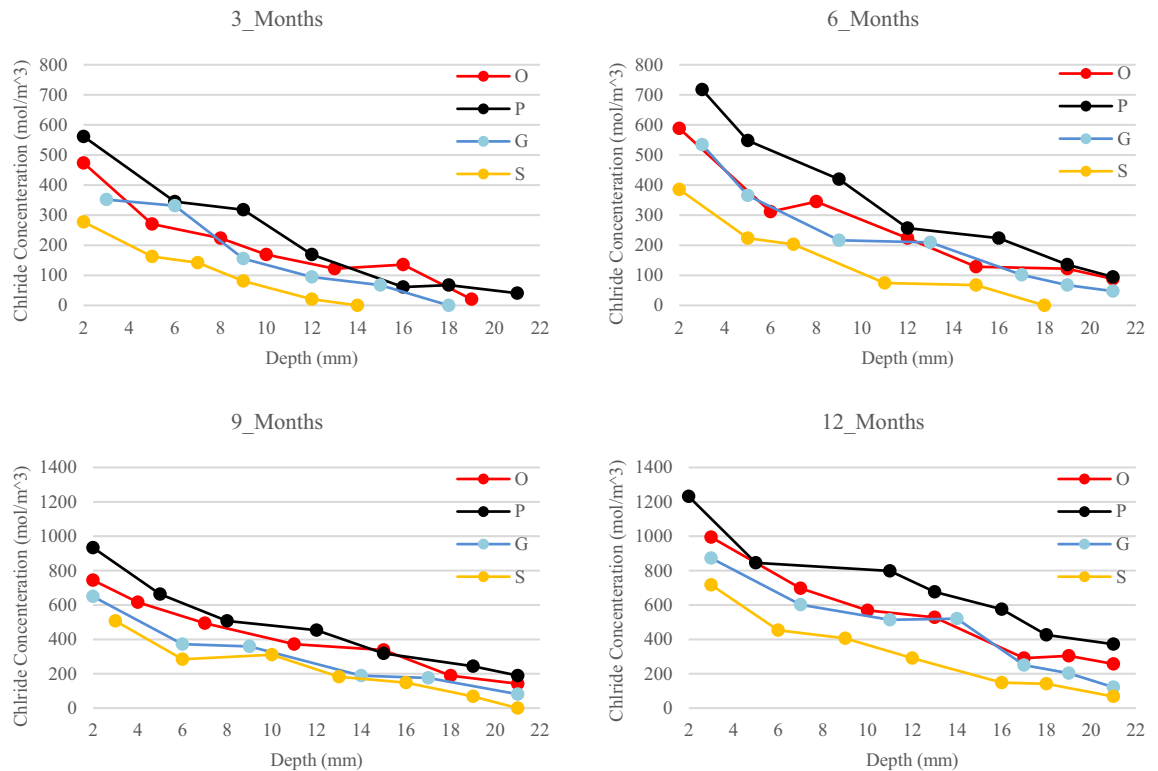


Fig. 4. Chloride ions penetration test results.

elements. Time steps for analysis were set at 60-day intervals thereafter. Following the simulation, various outputs, including chloride concentration profiles at different depths and concentration changes over time at specific locations (e.g., at the rebar), were generated. In this study, changes in chloride concentration along a horizontal line at the midpoint of the section were examined over time.

Experimental test and FE simulation results

Chloride ions penetration profile

The results of the penetration depth test of chloride ions, obtained after placing the samples in NaCl solution for 3, 6, 9, and 12 months, are shown in Fig. 4. At 3 months, the penetration depth of all samples, except for the P mixture, was observed to be less than 20 mm. The P mixture, with a w/b ratio of 0.55, exhibited higher chloride penetration compared to the O mixture with a w/b ratio of 0.45, indicating that a higher w/b ratio facilitates chloride ingress. At later ages, a similar trend was observed, with the SF-modified concrete (S) showing the lowest penetration depth at all four ages, followed by the GGBFS-modified concrete (G). These findings confirm that the incorporation of SCMs, specifically SF and GGBFS, substantially reduces the chloride diffusion coefficient and penetration depth compared to plain concrete. Our results align with those reported by Shittu et al.⁵⁶ and Basha et al.⁷⁰, who observed superior performance of SF over GGBFS in mitigating chloride ingress. Furthermore, Banar et al.⁵⁵ noted that the denser microstructure and refined pore system of SF concretes enhance resistance to chloride penetration, which supports our observations. Similarly, Liu et al.⁷¹ and Karein et al.⁷² also documented significant durability improvements and extended SL in SCM-modified concretes exposed to marine environments. Ramezani-pour et al.⁷³ highlighted that both SF and GGBFS effectively

Concrete Mixes	3 Months			6 Months			9 Months			12 Months		
	C_s (wt of conc.)	D ($\times 10^{-12} \text{ m}^2/\text{s}$)	R^2	C_s (wt of conc.)	D ($\times 10^{-12} \text{ m}^2/\text{s}$)	R^2	C_s (wt of conc.)	D ($\times 10^{-12} \text{ m}^2/\text{s}$)	R^2	C_s (wt of conc.)	D ($\times 10^{-12} \text{ m}^2/\text{s}$)	R^2
O	0.819	7.311	0.925	0.957	6.494	0.914	1.189	5.735	0.894	1.754	4.401	0.957
P	0.941	8.574	0.904	1.223	7.992	0.928	1.485	5.970	0.941	1.982	5.635	0.947
G	0.713	4.122	0.893	0.816	3.920	0.974	1.063	3.397	0.906	1.560	2.212	0.906
S	0.478	3.083	0.975	0.604	2.424	0.932	1.012	2.276	0.933	1.290	1.563	0.957

Table 3. Calculated chloride ions diffusion coefficient (D).

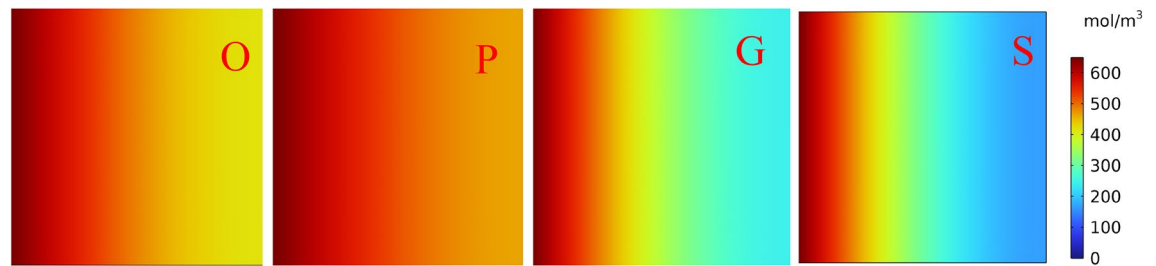


Fig. 5. Contour plots of ion concentration distribution (in mol/m^3) across the depth of concrete after 50 years of exposure to NaCl solution simulated with constant D_{3_months} .

inhibit chloride diffusion, though their efficiency depends on dosage and exposure conditions. Additionally, the duration of exposure significantly influenced chloride ingress; as shown in Fig. 4, samples exposed to NaCl solution for 12 months had penetration depths approximately 150% greater than those measured at 3 months.

The D values for various samples exposed to a NaCl solution for 3, 6, 9, and 12 months were computed using experimental data and MATLAB's Curve Fitting Toolbox, as reported in Table 3. The results indicate that the highest D, measured at 8.574, was observed for sample P after 3 months of exposure. Conversely, the lowest D, 1.563, was found for the sample containing SF after 12 months of exposure. These results demonstrate a clear trend: D decreases as the duration of exposure to the NaCl solution increases. This finding is consistent with the research conducted by Kazemian et al.⁷⁴, which observed a reduction in D by approximately 30 to 50% as the exposure duration increased from 119 days to 203 days. Additionally, the data presented in Table 3 show a decrease in the D for samples P, O, G, and S at each of the time points considered.

Table 3 presents the D calculated for concrete mixtures at various exposure durations (3, 6, 9, and 12 months). The results show a clear trend of decreasing diffusion coefficients over time, which is consistent with the expected ageing process of concrete, where the pore structure typically becomes denser as hydration continues. However, it is also noted that the surface chloride concentration increases significantly over the same period, more than doubling from 3 to 12 months. This observation warrants a deeper analysis, as it suggests a complex interaction between the diffusion process and the surface chloride concentration.

The increase in surface chloride concentration despite the decrease in diffusion coefficient may be due to several factors. As concrete ages, its pore structure refines, reducing the overall permeability and slowing down chloride ion ingress. However, the chloride ions that have already penetrated the surface may remain trapped near the surface or within the near-surface layers due to the reduced mobility of chloride ions in the denser microstructure. This could explain the increased concentration near the surface, as the material resists further chloride ingress while holding onto existing chloride ions.

Chloride ions penetration simulation with constant D

After determining the D based on laboratory results and performing FE simulations of chloride ion penetration within the CS, the concentration profiles throughout the structure closely resemble those illustrated in Fig. 5.

This figure represents the chloride ion concentration within the concrete after 50 years of exposure to a NaCl solution, calculated using a D_{3_months} . The results indicate that in samples O and P, which did not contain any SCMs, nearly the entire CS reached a chloride concentration of at least $500 \text{ mol}/\text{m}^3$. In contrast, concrete containing SF and GGBFS showed significantly lower chloride concentrations, with values below $200 \text{ mol}/\text{m}^3$ and $250 \text{ mol}/\text{m}^3$, respectively. Given the extensive number of figures required to depict chloride ion penetration depth at various ages, these data are presented as graphs in Fig. 6 for different concrete mixtures, based on a D_{3_months} .

In all subfigures of Fig. 6, the chloride concentration exhibits a decreasing trend with increasing depth from the concrete surface. This behaviour is anticipated, as chloride ions infiltrate the concrete from the surface, resulting in higher concentrations near the surface and progressively lower concentrations at greater depths. Over time, chloride concentration at any given depth tends to increase, as evidenced by the upward shift of the curves from 1 year to 50 years. This trend reflects the ongoing penetration of chloride ions into the concrete matrix over extended periods. Notably, the spacing between the curves diminishes over time, particularly after

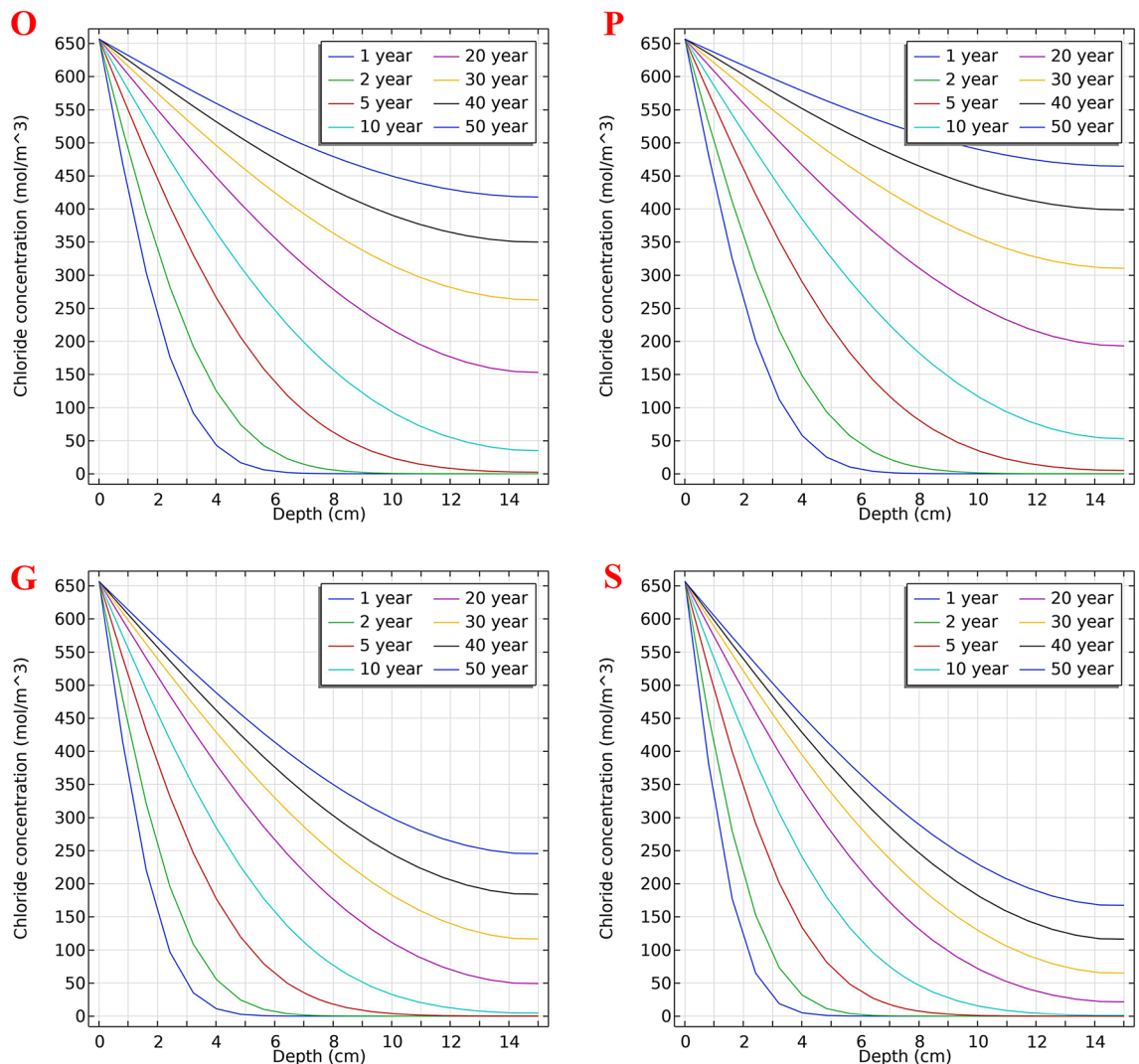


Fig. 6. Chloride ion concentration profiles at different depths in concrete over time, simulated for 50 years using COMSOL with constant D_{3_months} .

the 10-year mark, suggesting a deceleration in the rate of chloride ion penetration. This deceleration may be attributed to the densification of the concrete matrix or the reduction in available free chloride ions within the concrete. The P samples represent a condition characterised by relatively high initial chloride concentrations and penetration rates, resulting in elevated chloride concentrations at greater depths compared to other mixtures. Even after 50 years, the chloride concentration at a depth of 15 cm remains substantial, approximately 500 mol/m³. The O mixtures exhibit a slower initial rate of chloride penetration, as evidenced by the lower initial chloride concentrations relative to the P samples. The curves for the O mixtures are more closely spaced, indicating a slower overall penetration rate. However, the O mixtures still show slightly higher chloride concentrations at all depths and times compared to other scenarios, suggesting that this condition may have higher permeability or fewer protective measures against chloride ingress. In contrast, concrete mixtures containing SF demonstrate the lowest chloride concentrations across all times and depths. This suggests that the incorporation of SF as an SCM or the formation of a denser concrete matrix has effectively reduced chloride penetration. For in-depth comparison of chloride ingress across different concrete formulations and exposure durations, the FE simulation results of the penetration depth of chloride ions with constant D_{3_months} after 2, 5, 10 and 50 years are shown in Fig. 7.

In subfigure a, representing chloride concentrations after 2 years, it is evident that the initial chloride ingress is more pronounced in concrete with a higher w/b ratio (Sample P) compared to concrete with a lower w/b ratio (Sample O). The absence of SCMs in these samples leads to higher chloride concentrations, particularly near the surface. In contrast, the samples containing GGBFS (Sample G) and SF (Sample S) show significantly reduced chloride concentrations, with SF being the most effective in limiting chloride penetration. This early-stage performance highlights the significance of SCMs in improving the initial resistance of concrete to chloride ingress.

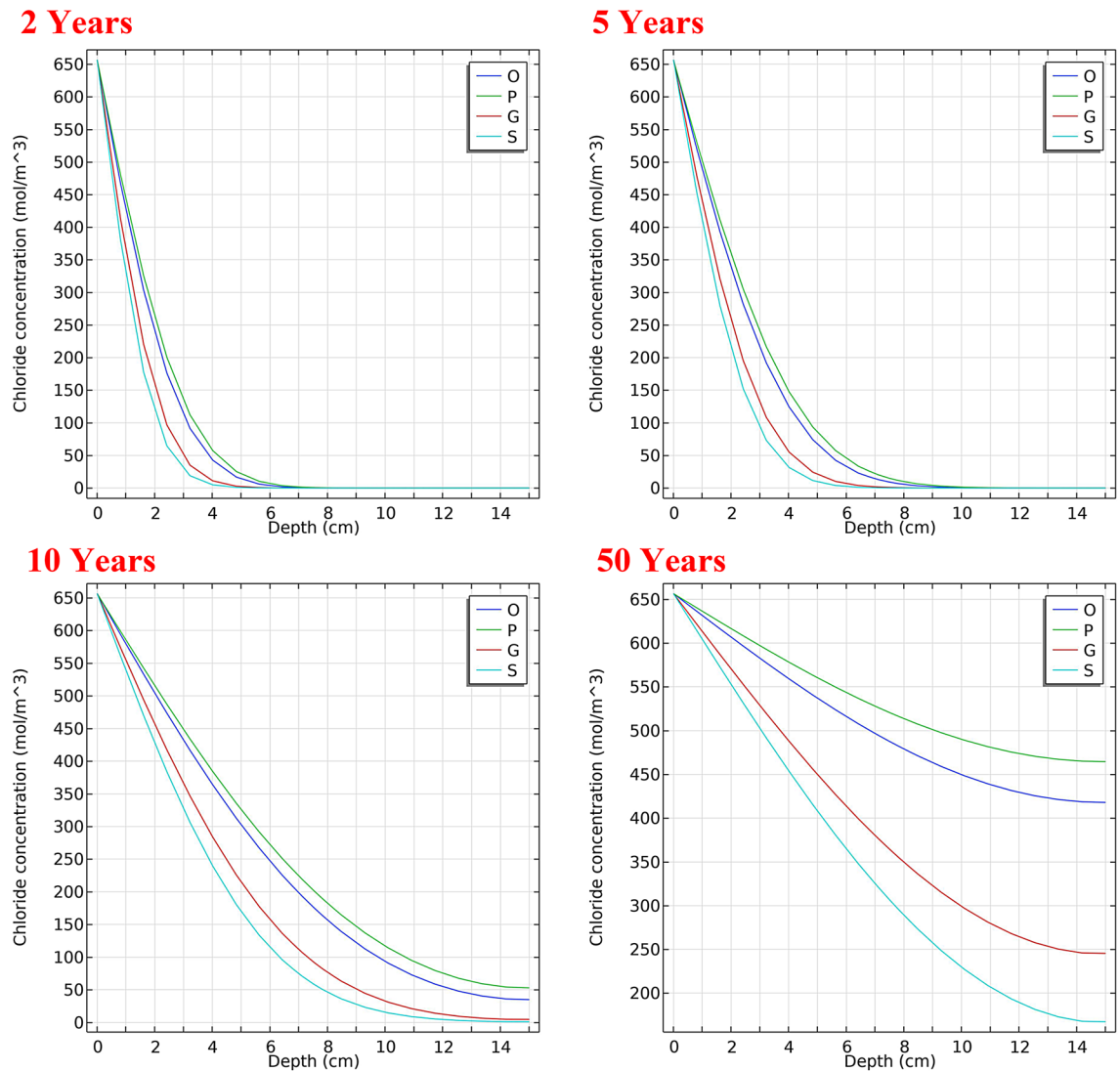


Fig. 7. Comparison of chloride ion concentration profiles at different depths in concrete after 2, 5, 10 and 50 years of exposure to NaCl solution simulated with constant D_{3_months} .

As shown in subfigure b, after 5 years of exposure, the chloride concentrations increase across all samples, but the rate of penetration begins to diverge more noticeably between the different concrete formulations. The higher w/b ratio in Sample P continues to facilitate greater chloride penetration compared to Sample O. However, the inclusion of GGBFS and especially SF results in significantly lower chloride concentrations at comparable depths, indicating that these materials play a crucial role in mitigating chloride ingress over time.

Subfigure c, depicting the chloride concentration profiles after 10 years, further highlights the long-term benefits of incorporating SCMs. Both Sample O and Sample P exhibit substantial increases in chloride concentrations at greater depths, with Sample P showing the most significant penetration. However, the concrete containing GGBFS maintains lower chloride levels, demonstrating the material's efficacy in reducing chloride diffusion. SF, once again, outperforms the other mixtures, maintaining the lowest chloride concentrations across all depths, even after a decade.

The trends observed in subfigure d, representing chloride concentrations after 50 years, provide critical insights into the long-term durability of the concrete samples. The chloride penetration continues to progress in all samples, but the extent of this penetration varies significantly. Sample P, with its higher w/b ratio, displays the most severe chloride ingress, reaching critical levels at depths that could lead to reinforcement corrosion. In contrast, both GGBFS and SF significantly reduce chloride penetration, with SF offering the best protection by maintaining the lowest chloride concentrations even after five decades.

The comparative analysis of these subfigures reveals the profound impact of the w/b ratio and the inclusion of SCMs on the chloride ingress and overall durability of concrete. The higher w/b ratio in Sample P consistently results in greater chloride penetration which expected because a higher w/b ratio generally leads to a more porous CS, allowing for easier chloride ingress, highlighting the importance of a lower w/b ratio in mitigating such risks. GGBFS is known to enhance the durability of concrete by refining the pore structure. The hydration

of GGBFS produces additional calcium silicate hydrate (C-S-H), which fills the capillary pores, reducing porosity and permeability. This refinement leads to reduced chloride ion ingress. Additionally, the pozzolanic reaction of GGBFS consumes calcium hydroxide ($\text{Ca}(\text{OH})_2$), further densifying the microstructure. In addition, GGBFS-modified concrete has a higher chloride binding capacity due to the formation of Friedel's salt, which binds free chlorides and reduces their mobility. This chemical binding reduces the concentration of free chloride ions that contribute to corrosion risk, as reflected by the lower chloride concentrations at deeper levels compared to non-SCM concrete. SF is characterised by its high pozzolanic activity and extremely fine particle size, which significantly enhances the concrete's microstructure. SF reacts with calcium hydroxide to form more C-S-H, leading to a denser and more impermeable matrix compared to GGBFS-modified concrete. Additionally, the ultra-fine particles of SF fill the micro-pores within the cement paste, effectively reducing chloride ion diffusion. In addition, the high surface area of SF leads to a rapid pozzolanic reaction, consuming CH more effectively than GGBFS and producing a more substantial amount of C-S-H. The combination of noted effects contributes to a more pronounced reduction in chloride ion diffusion and lower concentrations, which is consistent with the trends seen in Fig. 4.

Chloride ions penetration simulation with variable D

In the second part of the analysis, a FE simulation of chloride ion diffusion was conducted using a variable D . As presented in Fig. 8, the chloride ions penetration depth profile was evaluated at an age of 50 years, with a variable D_{3_months} .

The comparison between the simulations using a constant D and those employing a variable coefficient provides important insights into the durability of concrete under different conditions. The first simulation, based on a constant D , exhibits a more linear and uniform chloride penetration across various concrete types. In contrast, the second simulation, which incorporates a variable D , reveals a more complex interaction of diffusion rates, resulting in greater variability in both the depth and concentration of chloride ingress. For concrete with a w/b ratio of 0.45 without SCMs (as shown in subfigure a), the chloride concentration appears to penetrate more uniformly in the constant diffusion scenario. However, in the case of the variable D , the penetration is less regular, likely due to the varying diffusion rates within the concrete matrix. The inclusion of GGBFS, depicted in subfigure c, at a w/b ratio of 0.45, further reduces the penetration depth in the constant D simulation, with the colour gradient shifting more rapidly compared to concrete without SCMs. This reduction in penetration depth is even more pronounced in the variable D simulation, where chloride ingress is restricted to shallower depths, underscoring GGBFS's effectiveness in improving concrete's resistance to chloride ion diffusion. SF, as illustrated in subfigure d, provides the highest level of protection against chloride ion penetration among the materials studied. In both the constant and variable D simulations, chloride ingress is significantly limited, with a sharp transition from high to low concentrations. The superior performance of SF is attributed to its ability to refine the microstructure of the concrete, creating a highly dense matrix. This is particularly evident in the variable D simulation, where chloride penetration is almost completely inhibited, resulting in the shallowest penetration depth across all cases. Overall, the comparison between constant and variable D simulations highlights the critical role of lower w/b ratios and the inclusion of SCMs, such as GGBFS and SF, in mitigating chloride ion ingress. Although the variable D introduces a more dynamic and complex interaction between chloride ions and the concrete matrix, the general trends remain consistent: lower w/b ratios and the use of SCMs significantly enhance concrete's resistance to chloride penetration, thereby improving long-term durability. To further examine the trend of chloride ion penetration over time, Fig. 9 presents the results of the simulated chloride ion penetration profile, incorporating the variable D_{3_months} at different ages.

The comparison between simulations with a constant D (Fig. 6) and a variable D (Fig. 9) highlights the varying degrees of resistance to chloride ingress across different concrete mixes. As noted before, the first simulation assumes a constant D , providing a simplified model of chloride ingress, while the second simulation incorporates a variable D , allowing for a more realistic representation of concrete's dynamic properties over time. In the case of concrete without SCMs (subfigures a and b), the constant diffusion model predicts deeper chloride penetration compared to the variable diffusion model. For a concrete mix with a w/b ratio of 0.45 (subfigure a), the constant D suggests a chloride concentration of approximately 200 mol/m^3 at a 2 cm depth after 50 years, while the variable D model predicts a lower concentration of 150 mol/m^3 . Similarly, for the higher w/b ratio mix (0.55, subfigure b), the constant D model shows deeper penetration, with concentrations reaching around 300 mol/m^3 at 2 cm depth, compared to 250 mol/m^3 in the variable model. This suggests that

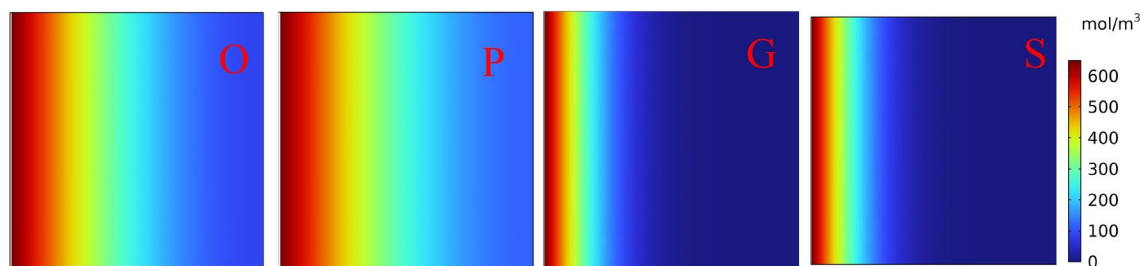


Fig. 8. Contour plots of ion concentration distribution (in mol/m^3) across the depth of concrete after 50 years of exposure to NaCl solution simulated with variable D_{3_months} .

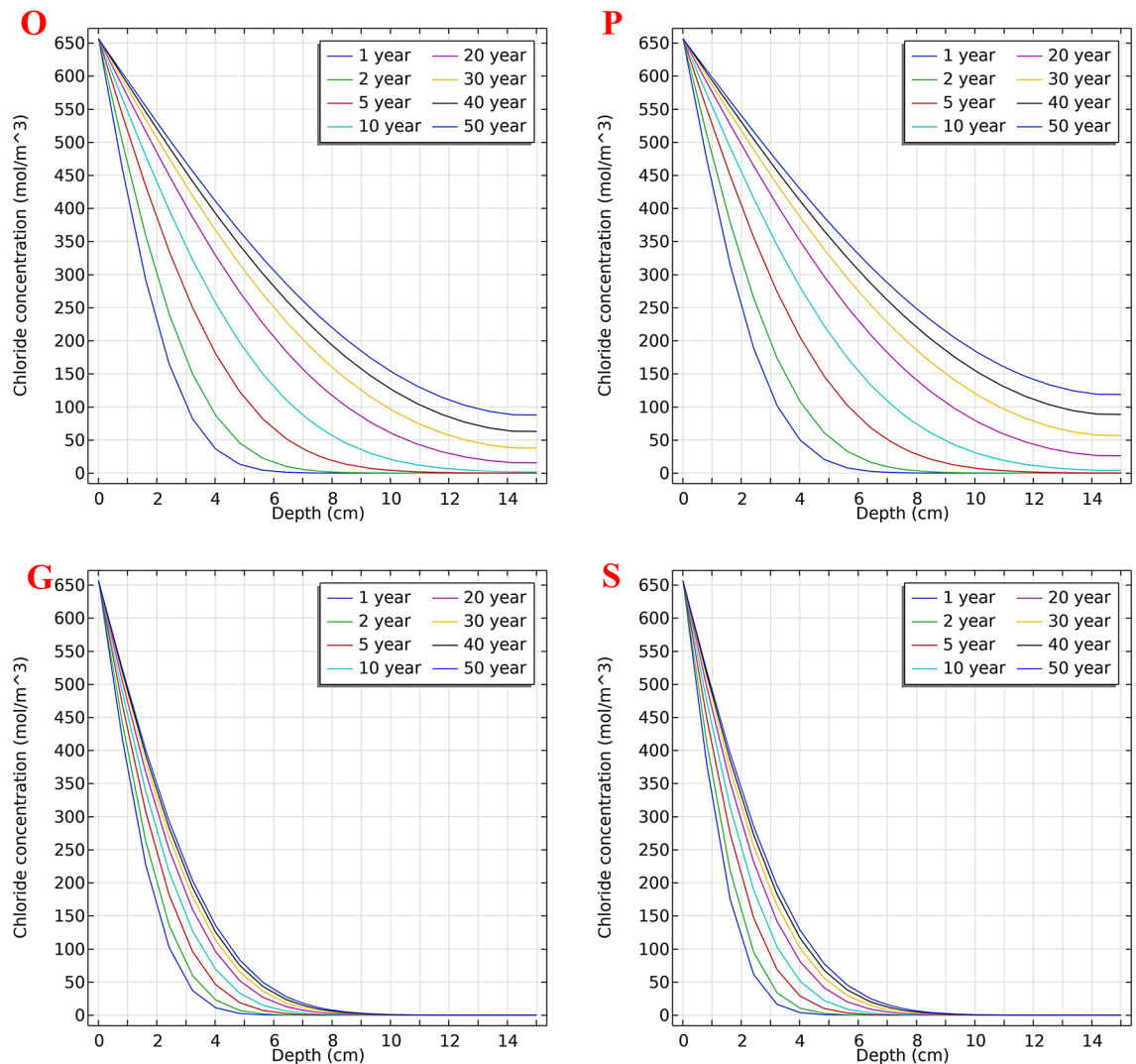


Fig. 9. Chloride ion concentration profiles at different depths in concrete over time, simulated for 50 years using COMSOL with variable D_{3_months} .

the variable diffusion model more effectively captures the slowing of chloride ingress due to long-term material property changes. For mixes containing GGBFS (subfigure c), both models demonstrate a significant reduction in chloride ingress compared to concrete without SCMs, but the variable diffusion model provides a more pronounced reduction in chloride concentration over time. In the constant D simulation, the concentration at a 2 cm depth after 50 years is around 150 mol/m³, whereas the variable diffusion model predicts a much lower concentration of 100 mol/m³. The penetration depth, where chloride concentration falls below 100 mol/m³, is reduced from approximately 10 cm in the constant diffusion model to 8 cm in the variable model. This highlights the GGBFS's ability to improve chloride resistance over time, which is more accurately captured in the variable D simulation. Concrete containing SF (subfigure d) shows the best resistance to chloride ingress in both models, with the variable diffusion model demonstrating superior performance. After 50 years, the chloride concentration at a 2 cm depth is around 50 mol/m³ in the variable model, compared to 100 mol/m³ in the constant model. The variable diffusion model further reduces the penetration depth where the chloride concentration falls below 100 mol/m³, from 8 cm in the constant model to 6 cm. SF, which significantly enhances the concrete's microstructure, offers greater resistance to chloride ion diffusion, and the variable model captures this performance improvement more effectively. In summary, the variable D model provides a more nuanced and accurate reflection of chloride ion penetration in concrete, especially when SCMs such as GGBFS and SF are incorporated. This model accounts for time-dependent changes in diffusion properties, which are not captured by the constant D model. The improved performance of SCM-containing concretes is more accurately reflected in the variable model, which better represents the long-term benefits of these materials in reducing chloride ingress and enhancing concrete durability.

Figure 10 presents the chloride ion concentration profiles for different concrete types subjected to NaCl exposure over periods of 2, 5, 10, and 50 years, simulated using a variable D in COMSOL. Across all exposure periods, concrete type P consistently displays the highest chloride concentrations at the surface, while concretes

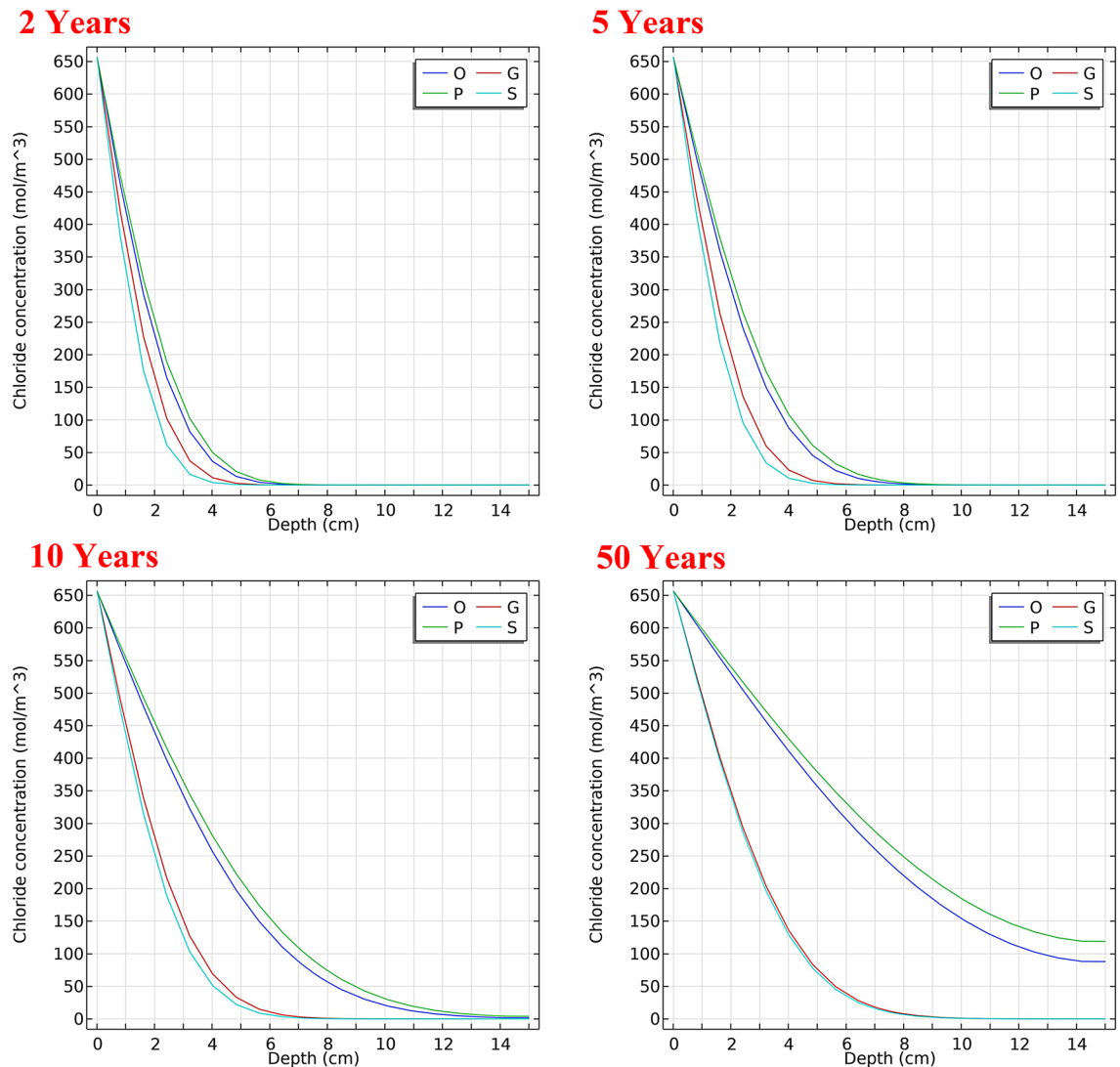


Fig. 10. Comparison of chloride ion concentration profiles at different depths in concrete after 2, 5, 10 and 50 years of exposure to NaCl solution simulated with variable D_{3_months} .

incorporating SCMs, particularly S and G, exhibit significantly lower chloride concentrations and steeper gradients. In the early stages of exposure (2 and 5 years), chloride ingress is more rapid for concretes without SCMs, with type P reaching a surface concentration of approximately 600 mol/m³. However, the presence of SCMs markedly reduces chloride penetration, with concrete containing SF maintaining the lowest chloride concentrations, even at shallow depths. Over longer exposure durations (10 and 50 years), the beneficial effects of SCMs become increasingly evident, as concrete types G and S continue to exhibit lower chloride concentrations at greater depths compared to types O and P. For instance, after 50 years, the chloride concentration at a depth of 10 cm is approximately 100 mol/m³ for type S, while type P exceeds 400 mol/m³. These findings underscore the superior long-term performance of SCM-containing concretes, particularly those incorporating SF, in limiting chloride ingress. The use of a variable D provides a more accurate representation of time-dependent chloride transport, highlighting the crucial role of SCMs in enhancing concrete durability under chloride exposure.

When comparing the constant D model (Fig. 7) to the variable D model (Fig. 10), several key differences emerge. While both models indicate a decrease in chloride concentration with depth, the variable D model predicts slower chloride ingress, especially at greater depths, underscoring the importance of accounting for time-dependent diffusion when modelling long-term chloride transport. In the 2-year exposure scenario, both models identify concrete type P as having the highest surface chloride concentration, followed by types O, G, and S. However, the variable D model predicts significantly lower chloride concentrations beyond 5 cm depth, with differences of up to 150 mol/m³ for concrete type P, suggesting that the constant model overestimates early penetration rates. As exposure time increases (subfigures b–d), the differences between the models become more pronounced. For example, after 5 years, chloride concentrations at 10 cm depth in concrete type P are 150–200 mol/m³ lower in the variable model compared to the constant model, indicating a more effective reduction in chloride transport over time. By 50 years, the variable model predicts concentrations up to 300 mol/

m³ lower than the constant model for concrete type P at 12 cm depth. The variable *D* model also more accurately captures the enhanced resistance offered by SCMs, particularly slag and SF, which exhibit significantly lower chloride penetration in this model. Overall, the variable *D* model provides a more realistic prediction of concrete performance under chloride exposure, making it essential for accurately assessing the durability and SL of CSs in chloride-laden environments.

Data validation

To evaluate the accuracy of the chloride ion penetration simulations in concrete, it is essential to compare the simulation results with experimental data. In this section, the average error is calculated as the average percentage difference between the concentration of chloride ions at each depth obtained from the simulations and those measured experimentally after 24 months, as shown in Table 4. This error provides a quantitative measure of how closely the simulation replicates the real-world behaviour of chloride diffusion into concrete. The average error is calculated for two diffusion models—one assuming a constant *D* and the other using a variable *D*—across four concrete mixtures (O, P, G, and S), representing different w/b ratios and the use of SCMs such as GGBFS and SF. By assessing the average error for each model and mixture, we can determine which model more accurately predicts chloride ion ingress over time and how various concrete compositions influence the simulation's reliability. For simulation with constant *D* across all mixtures, the average error remains significantly high, ranging from 48 to 75%. This indicates that the constant diffusion model does not accurately capture the experimental behaviour of chloride ion ingress over time. Mixtures with SCMs (G and S) show greater errors at certain times, particularly for the G mixture at 9 months (75%). This suggests that the constant *D* model struggles to represent time-dependent phenomena in mixtures containing SCMs. The errors are significantly lower in the variable *D* model across all time points and mixtures, with errors generally below 30%. This highlights the importance of accounting for time-dependent changes in the *D*, especially when simulating concrete with different w/b ratios and SCM. This table clearly demonstrates the importance of using a variable *D* in simulations, particularly for predicting chloride penetration in concretes with SCMs. It highlights the fact that concrete's resistance to chloride ions evolves over time, which must be captured in the simulation for more accurate SL predictions. Also, the average errors at simulation with *D*_{6-months} are remarkably low across all mixtures (12% for O, 18% for P, 13% for G, and 10% for S). This suggests that the variable *D* effectively captures the balance between the concrete's permeability and its resistance mechanisms during this intermediate stage of exposure. At simulation with *D*_{6-months}, the influence of SCMs like GGBFS and SF is particularly notable. SCMs enhance the long-term reduction in porosity, and by 6 months, their effects are well-developed. The variable diffusion model accurately reflects this, leading to low error values.

Service life prediction

SL prediction utilises validated models to estimate the durability of concrete based on a chloride threshold limit (0.4% by weight of the binder) and a predefined concrete cover (7.5 cm), with the results summarised in Table 5. The predicted SL for concrete specimens based on chloride ion penetration varies significantly depending on the assumed *D*, reflecting the influence of both the modelling approach and the evolving properties of the material. In this analysis, the *D* values were derived from experimental tests on concrete samples exposed to NaCl solution for 3, 6, 9, and 12 months, providing a realistic representation of how chloride ingress changes over time as the microstructure develops. This variability is influenced by the interaction between modelling assumptions, mixture composition, and age-dependent permeability, which collectively determine the concrete's resistance to chloride ingress^{75,76}. A comparison between constant and variable diffusion models clearly highlights these effects, particularly in SCM-containing concretes such as those with GGBFS and SF. For OPC concretes (O and P), the differences between the two models are moderate. For example, in concrete O, the constant *D* model estimated SLs increasing from 3.27 to 5.42 years as the exposure duration changed from 3 to 12 months, whereas the variable *D* model predicted slightly higher SLs, increasing from 6.00 to 6.83 years. This reflects the ability of the time-dependent model to account for the progressive densification of the microstructure due to ongoing hydration during prolonged NaCl exposure⁷⁷. In concrete P, the constant *D* model estimated shorter SLs, ranging from 2.79 to 4.23 years, while the variable *D* model predicted modest improvements, starting at 4.80 years and stabilising at 4.82 years by 12 months. This modest improvement is attributed to the limited resistance of OPC concrete with higher water content, which results in a more permeable matrix and reduced

Type of D	Mixtures	Average error (%)			
		3 months	6 months	9 months	12 months
Constant	O	55	56	60	63
	P	65	49	51	59
	G	57	63	75	52
	S	48	62	58	60
Variable	O	25	12	10	28
	P	23	18	26	22
	G	32	13	27	13
	S	20	10	30	12

Table 4. Data validation.

Type of D	Mixtures	Service life prediction (Year)			
		3 months	6 months	9 months	12 months
Constant	O	3.27	3.68	4.16	5.42
	P	2.79	3.00	4.00	4.23
	G	5.77	6.07	7.00	10.72
	S	7.71	9.79	10.42	15.14
Variable	O	6.00	5.29	5.31	6.83
	P	4.80	3.95	5.02	4.82
	G	202.00	48.04	30.23	62.13
	S	169.11	104.17	64.38	106.62

Table 5. Service life Prediction.

long-term performance^{48,49}. The use of SCMs significantly improved the SL predictions, especially when variable D values were considered. Concrete G demonstrated a marked improvement in SL under the variable diffusion model. While the constant D model estimated SLs between 5.77 and 10.72 years, the variable model dramatically increased these values, predicting an initial SL of 202 years at 3 months of exposure, decreasing to 62.13 years at 12 months. As observed, the predicted SL based on diffusion coefficients determined at lower reference ages, such as 3 months, yields substantially higher values compared to those at higher reference ages. This phenomenon can be explained by the nature of Eq. (3), in which the time-dependent diffusion coefficient at the target age t is calculated based on t_0 and the exponent m . When a smaller t_0 (e.g., 3 months) is chosen, the ratio $(t_0/t)^m$ becomes significantly lower than at larger t_0 , although the exponent $m < 1$ partially mitigates this effect. As a result, the computed diffusion coefficient at the target age becomes much smaller, leading to an unrealistically large predicted SL. Therefore, the remarkable increase in SL associated with lower reference ages is due to the exaggerated reduction in chloride ingress predicted at early ages, which explains the inverse trend observed in Table 5 for the time-dependent model compared to the constant model. In addition to this mathematical effect, the observed improvement is further supported by the progressive refinement of the pore structure and reduction in permeability provided by GGBFS under prolonged chloride exposure^{22,57,58}. Similarly, concrete S showed substantial differences between the two models. The constant diffusion model estimated relatively high SLs (7.71 to 15.14 years), reflecting the known benefits of SF in reducing chloride ingress. However, the variable diffusion model projected much higher SLs, from 169.11 years at 3 months to 106.62 years at 12 months. This significant enhancement is attributed to the ultra-fine particles of SF, which continue to refine the pore structure and densify the matrix over time, thereby greatly limiting chloride penetration^{52,55,56}. Similar trends have been reported in recent studies^{60–62}, which also demonstrated that the combined effects of SCMs and time-dependent parameters substantially enhance concrete durability and extend SL predictions in aggressive environments. These findings emphasise the necessity of adopting time-dependent diffusion models and incorporating mixture composition and microstructural evolution into SL predictions to achieve reliable estimates of long-term durability for reinforced concrete exposed to chlorides.

In light of the above findings on the significant influence of SCMs and time-dependent diffusion on SL of concrete, the following engineering recommendations are suggested. Incorporating SF or GGBFS as partial replacements for cement in concrete mixtures is strongly recommended for structures exposed to chloride-rich environments, such as those in marine or de-icing conditions. Specifically, using around 8% SF or 20% GGBFS by weight of binder, as investigated in this study, was shown to significantly reduce chloride ingress and improve the SL of reinforced concrete. Among the two SCMs, SF demonstrated superior performance due to its finer particle size and greater ability to densify the microstructure. Therefore, when maximum durability is desired, the use of SF is preferred, whereas GGBFS remains an effective alternative, particularly when longer setting times or improved workability are beneficial. These recommendations can guide engineers in selecting optimal mix designs to enhance the long-term performance of concrete in aggressive chloride-laden environments.

Conclusions

This study investigated chloride ion penetration in concrete using COMSOL Multiphysics simulations, focusing on mixtures with different w/b ratios and the incorporation of SCMs such as GGBFS and SF. SL predictions were performed using both constant and time-dependent diffusion coefficients, the latter derived from experimental tests conducted at 3, 6, 9, and 12 months. The predictive accuracy of these models was evaluated against 24-month experimental data. The following outlines the main findings of this study.

- The results clearly demonstrate that concrete durability against chloride ingress is strongly influenced by both the w/b ratio and the incorporation of SCMs. Concrete with a higher w/b ratio of 0.55 (P) exhibited significantly deeper chloride penetration compared to the 0.45 (O) mixture, indicating that reducing the w/b ratio improves impermeability. Furthermore, SCM-containing concretes—particularly those with SF (S)—showed the lowest penetration depths among all mixtures, highlighting the effectiveness of SCMs in refining the pore structure and substantially limiting chloride ingress.
- The SL predictions demonstrate the substantial influence of both SCMs and the choice of diffusion model. In mixtures without SCMs (O and P), the constant diffusion model estimated SLs of 2.79–5.42 years, which increased slightly to 4.80–6.83 years using the variable model, highlighting the moderate impact of time-de-

pendent modelling in OPC concretes. However, SCM-containing mixtures showed dramatic improvements under variable diffusion modelling. For GGBFS concrete (G), SL increased from 5.77 to 10.72 years (constant model) to 155–62.13 years (variable model), while SF concrete (S) improved from 7.71 to 15.14 years to 180–106.62 years. These results underscore the significant long-term benefits of SCMs, particularly when their effects on microstructure refinement and chloride resistance are captured through time-dependent modelling.

- The results of this study have important practical implications for the design and maintenance of concrete structures in chloride-laden environments. The use of SCMs, such as GGBFS and SF, not only enhances the long-term resistance of concrete to chloride ingress but also significantly extends SL, as clearly demonstrated by the variable diffusion model. Moreover, this model more accurately captures the evolving nature of concrete microstructure—particularly in SCM-containing mixtures—where chloride resistance improves continuously due to ongoing pore refinement. This behaviour is not reflected in the constant diffusion approach.
- The 24-month experimental results were used to validate the FE models developed using D values from 3, 6, 9, and 12 months. At this age, the variable diffusion model provided far more accurate chloride penetration predictions than the constant model, with errors below 30% compared to 48–75% in the latter. The best match was observed using 6-month D values (errors: O–12%, P–18%, G–13%, S–10%), highlighting the importance of time-dependent diffusion modelling.

In conclusion, this study underscores the vital role of SCMs such as GGBFS and SF in enhancing concrete resistance to chloride ingress and extending SL. The use of time-dependent diffusion coefficients led to more accurate and realistic durability predictions, especially for SCM-containing mixtures. The validated FE simulations confirmed the superiority of variable D models over constant ones. Future studies should incorporate long-term field data and additional environmental variables, such as temperature, moisture, and cracking, to further enhance the robustness and applicability of SL predictions.

Data availability

The datasets used and analysed during the current study available from the corresponding author on reasonable request.

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Author contributions

P.D.: Designed and developed the COMSOL simulation models, conducted experimental tests, analyzed the results, and wrote and edited the main manuscript text, including the introduction, methodology, and conclusion sections. R.B.: Contributed to the analysis of results and provided input on the interpretation of experimental data. F.M.: Responsible for the final review of the article, providing guidance on the structure, and ensuring consistency throughout the manuscript. Also served as the supervisor. A.A.R.: Supervised the research and provided critical insights during the entire study, particularly in the final manuscript review.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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